Proposed Solution & Innovation

SIMPLE TECHNOLOGY. MEANINGFUL RECOVERY.

Small Movements **Big Possibilities**

1 CORE SOLUTION



- Wearable glove combining flex sensors (finger movement) + surface EMG (muscle activation)
- ESP32 acquires both signals in real time during a prescribed hand exercise
- Lightweight classifier (Random Forest/SVM) flags each rep as correct or deviating
- Instant haptic/visual feedback + logged session data for the therapist to review

2 INNOVATION FACTOR

Same Capabilities.
Radically More Accessible.

RehabGrip		Commercial Sensing Gloves
₹6,500	vs	₹1L+
RehabGrip's BOM is ~₹5,500–6,500		Commercial sensing gloves cost ₹1L–3.5L+

- Not inventing multimodal sensing — flex+EMG fusion is validated in research
- Innovation is packing it into a consumer-cost, home-usable wearable
- First affordable product to combine movement + muscle-activation sensing with real-time feedback

3 KEY FEATURES

- Dual-sensor fusion: kinematic (flex) + muscle-effort (EMG) — neither alone catches compensation patterns
- Real-time rep-quality feedback (not just a rep counter)
- On-device signal processing, BLE streaming to companion dashboard
- Session logging + progress dashboard for therapist review between visits